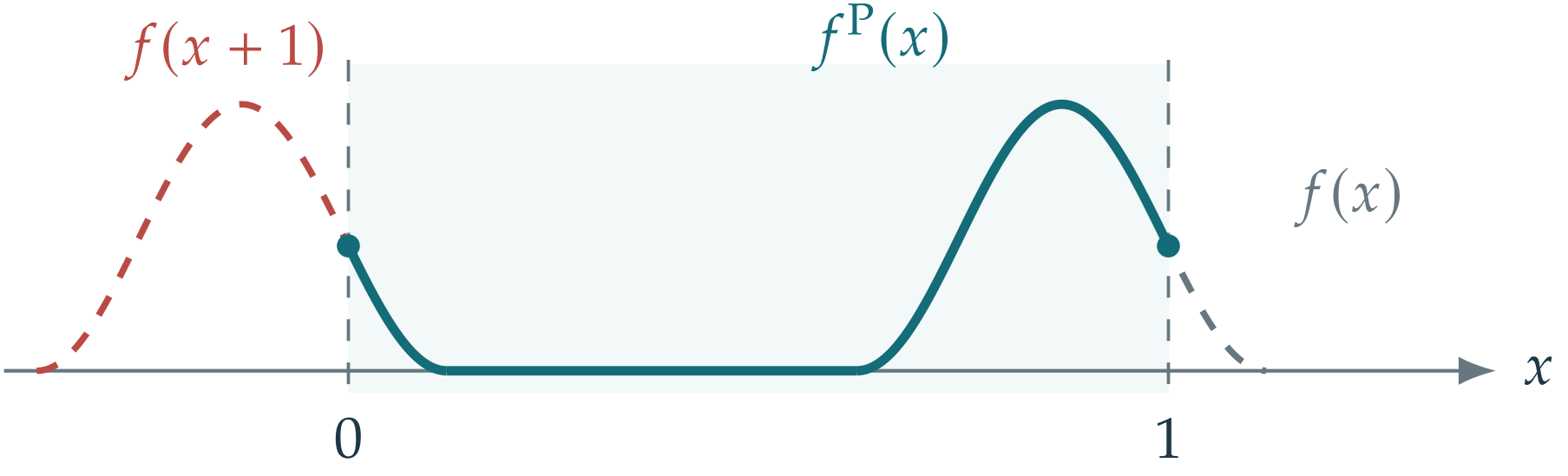

$$f^{\mathrm{P}}(x) = \sum_{k \in \mathbb{Z}} f(x-k), \quad f^{\mathrm{P}}(0) = f^{\mathrm{P}}(1)$$